

WESTERN MICHIGAN, MI, USA

WMO: 725394

Lat: 42.746N

Lon: 86.097W

Elev: 210

StdP: 98.83

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 04839

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.3	-18.9	0.7	-13.7	-16.8	0.9	-11.6	12.8	1.3	11.9	0.9	3.3	270	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.7	23.0	30.2	22.2	28.6	21.5	24.6	29.6	23.6	28.2	22.8	27.1	5.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	18.0	27.6	22.2	17.3	26.7	21.3	16.3	25.6	75.3	29.7	71.4	28.5	68.0	26.9	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.8	8.5	-19.2	33.8	3.2	1.6	-21.5	35.0	-23.4	36.0	-25.2	36.9	-27.5	38.1		
(5)			-19.6	26.0	3.1	1.2	-21.9	26.9	-23.7	27.6	-25.4	28.2	-27.7	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-3.2	-2.3	2.4	8.8	14.8	20.0	22.3	21.2	17.7	11.3	5.2	-0.1		
	DBStd	10.12	5.50	5.33	6.06	5.05	4.87	3.84	3.11	2.82	4.06	4.61	4.69	4.82		
	HDD10.0	1608	409	346	250	81	12	0	0	0	1	38	156	316		
	HDD18.3	3445	667	578	496	288	132	26	4	7	59	223	393	573		
	CDD10.0	1577	0	1	14	45	161	301	381	348	233	79	13	2		
	CDD18.3	373	0	0	1	3	23	76	126	97	41	6	0	0		
	CDH23.3	3101	0	0	8	26	208	633	1127	727	329	43	0	0		
	CDH26.7	853	0	0	1	3	55	187	348	168	84	6	0	0		
(14) Wind	WSAvg	3.8	4.6	4.4	4.2	4.4	3.8	3.3	3.0	2.7	3.0	3.7	4.3	4.5		
(15) Precipitation	PrecAvg	898	54	45	54	86	105	85	66	82	84	96	72	71		
	PrecMax	1320	105	87	101	182	237	227	147	172	276	226	205	194		
	PrecMin	686	17	18	17	19	32	22	9	25	9	14	15	34		
	PrecStd	171	26	18	29	43	51	59	33	43	60	56	43	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	15.5	23.8	26.7	31.1	32.8	34.1	32.3	31.7	27.5	20.5	15.3		
		MCWB	10.8	12.4	17.2	17.3	21.3	23.2	24.5	23.6	21.4	20.5	14.0	13.1		
	2%	DB	8.9	11.1	18.8	22.9	28.0	30.9	32.0	30.2	29.0	24.0	17.2	12.1		
		MCWB	7.1	8.2	14.3	15.1	19.7	22.2	23.7	22.7	21.7	18.3	13.1	10.5		
	5%	DB	6.2	7.2	15.0	20.3	25.6	28.8	30.2	28.4	27.1	21.6	14.8	8.8		
		MCWB	4.2	4.6	10.9	13.2	18.4	21.2	22.6	21.8	20.3	17.2	11.1	6.9		
	10%	DB	3.9	4.4	11.3	17.5	23.0	27.2	28.5	27.2	25.0	19.0	12.6	6.3		
		MCWB	2.5	2.6	7.6	11.7	16.9	20.7	21.9	21.2	19.4	15.3	9.5	4.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	12.6	17.4	18.9	22.7	25.0	26.1	25.2	24.1	21.6	15.6	13.7		
		MCDB	12.7	14.1	23.4	24.6	28.8	30.5	31.5	30.1	28.8	26.2	18.1	15.1		
	2%	WB	7.5	7.9	14.8	16.2	20.9	23.5	25.0	24.0	22.7	19.5	13.7	10.3		
		MCDB	8.4	10.8	18.3	21.2	26.2	28.9	30.2	28.2	26.8	22.5	16.7	11.8		
	5%	WB	4.4	4.8	11.6	14.2	19.5	22.4	23.9	23.1	21.5	17.7	11.7	7.0		
		MCDB	5.9	6.9	14.4	18.9	24.2	27.3	28.7	26.9	25.3	20.4	14.3	8.4		
	10%	WB	2.6	2.8	7.8	12.1	18.1	21.4	22.9	22.2	20.3	15.6	9.7	4.5		
		MCDB	4.0	4.4	10.8	16.7	22.2	25.8	27.3	25.9	24.1	19.0	12.5	6.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.2	7.0	8.5	10.2	10.5	10.1	10.3	10.2	10.9	9.3	7.2	5.6		
		MCDBR	8.6	10.0	12.4	14.5	12.8	11.9	11.8	11.8	12.7	12.6	10.6	8.7		
	5% WB	MCWBR	7.4	7.8	8.5	8.5	7.1	5.8	5.5	5.9	6.4	7.7	7.7	7.8		
		MCWBR	7.9	9.6	11.6	12.4	11.5	10.2	10.1	9.3	10.3	11.0	9.6	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.339	0.344	0.379	0.411	0.428	0.425	0.424	0.393	0.353	0.321	0.301		
		taud	2.403	2.269	2.365	2.297	2.268	2.272	2.281	2.283	2.341	2.442	2.477	2.445		
	Ebn at Noon		837	850	898	887	864	848	846	834	835	830	809	804		
		Edh at Noon	78	105	108	125	132	132	130	126	110	88	73	69		
(44) All-Sky Solar Radiation	RadAvg		1.23	2.04	3.54	4.74	5.51	6.22	6.33	5.40	4.31	2.68	1.58	1.01		
	RadStd		0.13	0.28	0.33	0.43	0.51	0.41	0.28	0.31	0.38	0.31	0.24	0.14		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page